

Roadmap Document

Tentative Project Title: Natural Language Multi-Drone Mission Planning and Coordination using Shepherd-AI

Project Goal: Build a research prototype where an operator can issue voice or text commands such as:

“Send two drones north and scan the crops while another drone checks the irrigation canal.”

The system should:

- Convert speech to text.
- Extract mission intent.
- Ground vague commands into map locations.
- Plan executable drone tasks.
- Coordinate multiple virtual drones.
- Analyze images using computer vision.
- Validate safety constraints.
- Generate mission feedback and reports.

The entire project is implemented as a software simulation using Python and publicly available datasets, so no physical drone hardware is required.

Technology Stack:

- **Development Platforms (choose any):**
 - Google Colab
 - Jupyter Notebook
 - Visual Studio Code (VS Code)
- **Programming Language:** Python
- **Version Control:** Git and GitHub
- **Natural Language Processing:** Hugging Face Transformers, spaCy
- **Speech Recognition:** Whisper
- **Computer Vision:** Ultralytics YOLO
- **Planning & Graphs:** NetworkX
- **Visualization:** Matplotlib, Folium
- **Data Processing:** Pandas, NumPy

Suggested Google Drive Structure (Google Colab)

ShepherdAI_Project/

```
|
|└─ notebooks/
|  |└─ Notebook1_Setup.ipynb
|  |└─ Notebook2_NLP.ipynb
|  |└─ Notebook3_Grounding.ipynb
|  |└─ Notebook4_Planner.ipynb
|  |└─ Notebook5_Scheduler.ipynb
|  |└─ Notebook6_Vision.ipynb
|  |└─ Notebook7_Safety.ipynb
|  └─ Notebook8_FinalDemo.ipynb
|  └─ Notebook9_Evaluation.ipynb
|
|└─ datasets/
|  |└─ maps/
|  |└─ aerial_images/
|  |└─ sample_audio/
|  └─ commands/
|
|└─ outputs/
|└─ reports/
└─ src/
```

Week 1 – Environment Setup and Repository Exploration

Learning Goals:

- Understand the Shepherd-AI project.
- Set up the Google Colab environment and Organize project files.

Tasks

1. Create a Google Drive folder for the project.
2. Open a new notebook.
3. Clone the Shepherd-AI repository:

```
! git clone https://github.com/cyberuniversal/shepherd-ai.git
```
4. Explore the repository structure and README.
5. Install required Python packages.
6. Document the architecture with a simple flow diagram.
7. Create a GitHub repository for your own extensions.

Deliverables

- Working Colab environment.
- Repository exploration notes.
- Architecture sketch.
- Personal project journal.

Week 2 – Speech Recognition and Intent Extraction

Learning Goals: Convert spoken or typed commands into structured mission intents.

Tasks

1. Create a notebook for NLP.
2. Accept typed commands and uploaded audio.
3. Convert speech to text using Whisper.
4. Extract:
 - Action
 - Number of drones
 - Location
 - Target
 - Constraints
5. Store results in JSON format.

Example:

Students can create their own dataset by recording commands such as:

- "Send one drone north."
- "Inspect the greenhouse."
- "Scan the crops."
- "Return all drones."
- "Check the irrigation canal."

Save recordings in WAV format and maintain a corresponding transcript file.

Example:

audio_001.wav

→ Send two drones north and inspect the crops.

audio_002.wav

→ Scan the western field.

audio_003.wav

→ Return drone three.

One Such Example:**Input:**

"Send two drones north and inspect the crops."

Output:

```
{  
  "action": "inspect",  
  "count": 2,  
  "location": "north",  
  "target": "crops"  
}
```

Deliverables

- Speech-to-text pipeline.
- Intent parser.
- Example command dataset.

Week 3 – Command Grounding and Map Representation

Learning Goals: Translate human directions into executable coordinates.

Tasks

1. Build a simple map of a virtual farm or campus.
2. Create a CSV with regions and coordinates.
3. Map terms like “north” or “greenhouse” to those coordinates.
4. Display locations with Folium.
5. Test grounding using multiple example commands.

Sample Dataset:

region, latitude, longitude

north_field,17.5000,78.5000

south_field,17.4950,78.5000

east_greenhouse,17.4970,78.5050

west_storage,17.4970,78.4950

Deliverables

- Map dataset.
- Grounding module.
- Interactive map visualization.

Week 4 – Mission Planning

Learning Goals: Convert high-level requests into executable task sequences.

Tasks

For a command such as: “Scan the north field.”

Generate:

1. Take off.
2. Fly to destination.
3. Capture images.
4. Run vision model.
5. Save results.
6. Return.

Represent tasks using lists or graphs.

Deliverables

- Planner notebook.
- Task decomposition examples.
- Mission flow diagrams.

Week 5 – Multi-Drone Coordination

Learning Goals: Allocate tasks across multiple virtual drones.

Tasks

1. Simulate three drones.
2. Automatically assign missions.
3. Display assignments in tables.
4. Implement simple scheduling logic.
5. Compare assignment strategies.

Deliverables

- Scheduler notebook.
- Drone allocation visualization.
- Example simulations.

Week 6 – Computer Vision Integration

Learning Goals: Analyze drone imagery automatically.

Tasks

1. Install Ultralytics YOLO in Colab.
2. Download and organize public datasets.
3. Run inference on aerial images.
4. Save detections and confidence values.
5. Generate annotated outputs.

Recommended Public Datasets

Purpose	Dataset	Links
Drone object detection	VisDrone	https://aiskyeye.com/download/
UAV detection and tracking	UAVDT	https://sites.google.com/view/dataset4uavdt
Aerial imagery	DOTA	https://captain-whu.github.io/DOTA/
Satellite imagery	xView	https://xviewdataset.org/
Agricultural inspection	Agriculture-Vision	https://www.agriculture-vision.com/
Speech testing	Self-recorded WAV files	Students can create their own dataset by recording commands
Mission locations	Custom CSV or GeoJSON	Create a CSV file describing mission regions.

Deliverables

- Vision notebook.
- Detection examples.
- Annotated images.
- Detection summary table.

Week 7 – Feedback, Safety, and Integration

Learning Goals: Enable interaction and mission validation.

Tasks

1. Build a clarification dialogue.
2. Simulate mission status updates.
3. Add safety checks:
 - Battery threshold
 - Restricted areas
 - Altitude limits
 - Drone availability
4. Integrate previous modules into one workflow.

Deliverables

- Feedback module.
- Safety validator.
- Integrated prototype.

Week 8 – End-to-End Demonstration and Experimental Evaluation

Learning Goals: Evaluate the complete system.

Tasks

Execute a full scenario:

“Send two drones north to inspect crops and one drone east to inspect irrigation.”

The system should:

1. Convert speech to text.
2. Extract mission intent.
3. Ground commands into locations.
4. Generate a mission plan.
5. Assign drones.
6. Validate safety.
7. Process imagery.
8. Produce a mission report.

Measure:

- Intent extraction accuracy.
- Grounding accuracy.
- Scheduling quality.
- Detection performance.
- Overall execution time.

Deliverables

- Complete working prototype.
- Evaluation results.
- Demonstration screenshots and logs.
- Organized source code repository.

Week 9 – Research Paper Draft Preparation

Learning Goals: Transform the completed project into a scientific manuscript.

Tasks

Write the first draft of:

1. Title
2. Abstract and Keywords
3. Introduction
4. Motivation
5. Problem Statement
6. Objectives
7. Related Work
8. Proposed Method
9. System Architecture
10. Methodology

Prepare:

- Architecture figures
- Workflow diagrams
- Tables describing datasets and modules
- Citations and references

Deliverables

- First research paper draft.
- System figures and tables.
- Organized bibliography.

Week 10 – Research Paper Finalization and Presentation

Learning Goals: Finalize the manuscript and prepare project dissemination materials.

Tasks

Complete and refine:

- Experimental Setup
- Datasets
- Evaluation Metrics

- Results and Analysis
- Discussion
- Limitations
- Future Work
- Conclusion

Additionally:

- Proofread the manuscript.
- Format references consistently.
- Create presentation slides.
- Prepare a project demonstration.
- Update the README with installation and usage instructions.

Deliverables

- Final research paper.
- Presentation slides.
- Complete documentation.
- Source code package.
- Demo materials suitable for competitions or student conferences.

Suggested Notebook Organization

1. Notebook 1 – Environment Setup and Shepherd-AI Exploration
2. Notebook 2 – Speech Recognition and NLP
3. Notebook 3 – Command Grounding
4. Notebook 4 – Mission Planning
5. Notebook 5 – Multi-Drone Scheduling
6. Notebook 6 – Computer Vision
7. Notebook 7 – Safety and Feedback
8. Notebook 8 – End-to-End Integrated Demo
9. Notebook 9 – Evaluation, Visualizations, and Research Figures

Expected Final Outcome

At the end, the student will have:

- A functional software prototype implementing natural-language-controlled multi-drone mission planning.
- Modules for speech recognition, intent extraction, command grounding, mission planning, multi-drone scheduling, safety validation, and computer vision.
- Experimental results supported by visualizations and evaluation metrics.
- A professionally structured research paper with figures, tables, references, and discussion.
- Presentation materials and documentation suitable for science fairs, research competitions, or introductory academic publication.

References:

Natural Language for Robotics

1. "Code as Policies: Language Model Programs for Embodied Control"
<https://arxiv.org/abs/2209.07753>
2. "SayCan: Grounding Large Language Models for Robotic Control"
<https://arxiv.org/abs/2204.01691>
3. "RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control"
<https://arxiv.org/abs/2307.15818>

Multi-Agent and Planning

4. "Voyager: An Open-Ended Embodied Agent with Large Language Models"
<https://arxiv.org/abs/2305.16291>
5. "Generative Agents: Interactive Simulacra of Human Behavior"
<https://arxiv.org/abs/2304.03442>

Drone Vision and Perception

6. "Detection and Tracking Meet Drones Challenge" (VisDrone benchmark)
 - Refer to the official VisDrone resources:
 - [VisDrone GitHub Repository](#)
 - [VisDrone Official Downloads](#)
7. "VisDrone-DET2019: The Vision Meets Drone Object Detection Challenge"
https://openaccess.thecvf.com/content_ICCVW_2019/papers/VISDrone/Du_VisDrone-DET2019_The_Vision_Meets_Drone_Object_Detection_in_Image_Challenge_ICCVW_2019_paper.pdf